

RAD Model (Rapid Application Development)

Introduction

Good morning everyone. Today I am going to explain the RAD Model and Agile Model. First, let us understand RAD, which stands for Rapid Application Development.

RAD is a software development approach that mainly focuses on developing software quickly. Instead of spending too much time in planning, RAD uses rapid prototyping and repeated improvements. This helps companies deliver software faster.

Main Features of RAD

1. Prototyping

The first feature is prototyping. In RAD, developers create working models at an early stage. These models help users understand how the software will work. Because of this, requirements can be checked early and feedback can be collected quickly.

2. User Involvement

The second feature is user involvement. Customers continuously participate during development. Their feedback helps developers understand user needs better and create more useful software.

3. Component Reuse

Another important feature is component reuse. Instead of creating everything from the beginning, developers reuse existing modules and libraries. This saves both time and cost.

4. Fast Delivery

RAD also focuses on fast delivery. Since reusable components and prototypes are used, software can be developed in shorter cycles and delivered much faster.

Advantages of RAD

Some major advantages of RAD are:

- Faster software development
- Better customer feedback

- More flexibility during development
 - Suitable for projects with clear requirements
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Why Agile Emerged

Now let us understand why Agile became important.

Traditional software models had many problems. They followed rigid processes, took a long time to develop software, and could not easily handle changing requirements. Customers also received feedback very late.

Because of these problems, Agile emerged.

Dynamic Requirements

Customer requirements changed frequently, making long-term planning difficult.

Fast-Changing Markets

Companies needed faster releases to stay competitive.

Frequent Releases

Businesses wanted software updates regularly instead of waiting until project completion.

Customer Collaboration

Customers wanted continuous involvement throughout development.

Agile Model

Agile was introduced through the Agile Manifesto in 2001. It focuses on collaboration, flexibility, and continuous software delivery.

Key Principles

1. Flexibility

Agile accepts changing requirements and adapts quickly.

2. Faster Delivery

Software is delivered in smaller parts so customers receive value earlier.

3. Continuous Improvement

Teams regularly review their work and improve their processes.

Agile Frameworks

Scrum

Scrum uses short development cycles called sprints, usually lasting 2 to 4 weeks. Teams maintain a product backlog and conduct daily standup meetings.

Kanban

Kanban uses visual boards where tasks move through different stages. It limits work in progress and supports continuous workflow.

Extreme Programming (XP)

XP focuses on improving software quality using pair programming, test-driven development, and continuous integration.

Conclusion

To conclude, RAD focuses mainly on speed and rapid development, while Agile focuses on flexibility and continuous improvement. Both approaches help organizations build better software faster and more efficiently.

Thank you.